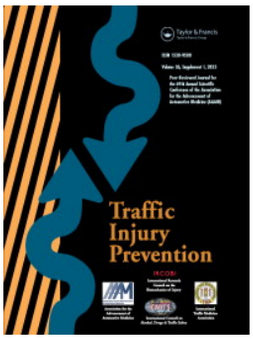

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A mechanistic approach to modeling omnidirectional motorcyclist injury risk

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ABSTRACT

Objective: As of 2022, motorcyclist fatalities in the United States had risen 38% since 2010, representing 15% of all U.S. traffic fatalities. Recently developed injury risk models have sought to better predict injury potential for certain collision configurations involving motorcycles using relative speed as a primary predictor variable. Advancing the state-of-the-art, this study developed injury risk models for motorcyclist collisions with passenger vehicles across all planar configurations and incorporated biomechanically-relevant predictor variables including a novel speed parameter.

Methods: We analyzed real-world crash data from the German In-Depth Accident Study (GIDAS) (1999–2023) to examine motorcyclist injury patterns and create injury risk functions at the MAIS2+F, 3+F, 4+F, and 5+F levels. Biomechanically relevant variables, including age (*via* a spline function), sex, and a geometric-based assessment of motorcyclist post-impact response (i.e., potential for a normal projection), were considered. *Effective Collision Speed*, combining passenger vehicle and motorcycle speeds while accounting for reduced engagement associated with frictional effects in side impacts, was employed as an important predictor. We analyzed the impact of reweighting the dataset to German national statistics, addressing GIDAS' bias toward severe and fatal collisions.

Results: The dataset comprised 2,499 passenger-vehicle to motorcycle collisions, of which 59% involved contact with the front of the passenger vehicle, 25% the side, and 16% the rear. 37% of motorcyclists sustained AIS2+F injuries and 11% sustained AIS3+F injuries. At the MAIS3+F level, the lower extremities were the most commonly injured body region, followed by the thorax and head. Age significantly influenced injury risk at MAIS2+F and MAIS3+F levels. A potential normal projection was associated with higher injury risk, significant only for MAIS2+F. *Effective Collision Speed* emerged as the sole significant predictor for higher severity levels.

Conclusions: These findings highlight the importance of incorporating biomechanical factors and refined speed metrics into motorcyclist injury risk models. The proposed *Effective Collision Speed* demonstrated strong predictive capability, offering a more comprehensive approach for assessing injury potential across varied crash configurations.

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Motorcyclist; motorcycle; injury risk; logistic regression; GIDAS; passenger vehicles

Introduction

Motorcycle usage has been increasing in the U.S., prompting renewed interest in understanding the factors contributing to motorcyclist injury risk. In the United States, the number of registered motorcycles has more than doubled between 2002 and 2023 – a 102% increase – with an average annual growth rate of 3.5% (Teoh 2023). Similarly, Germany has experienced an annual growth of 2.1% in motorcycle registrations from 2009 to 2024 (DESTATIS 2025). Despite their still relatively small numbers compared to passenger vehicles in these countries, motorcyclists face a disproportionate risk on the road. In 2022, motorcyclists accounted for 15% of all traffic fatalities in the United States, with the fatality rate per 100 million vehicle miles traveled (26.16) being nearly 22 times higher than that for passenger car occupants (1.20) (NHTSA 2024a). Notably, motorcyclist fatalities in the U.S. have risen 38% since 2010, with 15% of fatal motorcycle crashes involving collisions with passenger cars. Young riders, particularly those aged 21 to 24, experienced a 16% increase in fatalities from 2021 to 2022. The number of injured motorcyclists is correspondingly high, with approximately 218,000 treated in hospitals in 2022 (National Center for Injury Prevention and Control and CDC 2025). The situation is similarly concerning in Germany, where in 2023, motorcyclists comprised 18% of all traffic fatalities and 15% of the seriously injured (DESTATIS 2024). These trends underscore the need for precise and context-specific injury risk models for motorcyclists. Understanding the factors that elevate the risk of serious injury or fatality in motorcycle crashes is crucial for developing targeted safety interventions and informing policy. This paper aims to enhance our understanding of

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motorcyclist injury risk by developing models that account for the diverse crash constellations observed in collisions between motorcycles and passenger vehicles. Therefore, we extended the methodology introduced by Schubert et al. (2024) for bicycle collisions, adapting it to motorcycles. Thus, we employed the concept of *Effective Collision Speed* to capture the physically relevant influence of impact location on modeled speed. Additionally, we incorporated geometric factors that determine the likelihood of a normal projection - extending the notion of forward projection in frontal collisions to all impact orientations.

Existing injury risk models

Motorcyclists face a significantly higher risk of injury and fatality compared to occupants of other vehicles. Understanding the factors that contribute to this risk is crucial for developing effective safety interventions. Several recent investigations have developed injury risk models for motorcyclists using logistic regression and a variety of predictor variables, crash scenarios, and data sources.

Three data sources have primarily been used: the Motorcycle Accidents In Depth Study (MAIDS) (ACEM 2003), the 2016 Motorcycle Crash Causation Study (MCCS), and the German In-Depth Accident Study (GIDAS). MAIDS was a European in-depth study of motorcycle accidents in five different countries over three years collecting 921 crashes. MCCS (Nazemetz et al. 2019) includes data from 351 crash investigations in the United States. In contrast, GIDAS, which has been collecting crash data since 1999, and continues to do so, recorded 42,647 fully documented and reconstructed crashes until 2023, of which 6,157 involved motorized two-wheelers (FAT et al. 2023).

In addition to different data sources, studies have used varying definitions of motorcycles. According to UNECE (2017), motorized two-wheelers are classified into categories L1 and L3, based on engine cylinder capacity and maximum operating speed. Category L1 includes vehicles with an engine capacity below 50 cm^3 and a maximum speed not exceeding 50 kph, whereas category L3 comprises vehicles that exceed at least one of these thresholds. Teoh (2023) reports a rise in scooter registrations, often classified as L1, in the U.S. To provide a comprehensive overview, this study includes both categories.

MAIDS provided foundational insights into motorcycle injury risk, particularly regarding fatal outcomes (Smith 2009). In MAIDS both L1 and L3 motorcycle cases were examined, with a notable finding that approximately 25% of fatalities involved L1 motorcycles, with 12% of the fatalities involving motorcyclists under 18 years. 84% of the crashes involved another vehicle. Their multivariate logistic regression identified increased motorcycle crash speed and rider age as significant predictors of fatality risk.

To date, studies using the GIDAS database have focused exclusively on L3 motorcycles (Otte et al. 2015; Ding et al. 2019; Lubbe et al. 2022). Otte et al. (2015) examined categorical rider age, body mass index (BMI), collision constellation, collision partner type, and both the motorcycle's driving speed and the relative speed at collision. Their analysis was limited to a subset of L3 motorcycles (engine capacity > 125 cm^3) and riders younger than 18 were not considered. This age exclusion is notable given that data from DESTATIS (2020) indicate that nearly 20% of fatalities and injured motorcyclists are under 18. Thus, the Otte et al. (2015) model may not fully capture the risk profiles of younger motorcyclists. Ding et al. (2019) considered relative speed, ride pre-crash stability, collision partner type, motorcycle impact location, and the rider's impact mechanism in their injury risk functions. Cases involving motorcycle rear impact, run-overs, or unhelmeted riders were excluded from their analysis. Ding et al. (2019) concluded that a model based on relative speed provided the best proxy for crash severity, outperforming delta-v based models. More recently, Lubbe et al. (2022) focused exclusively on frontal collisions with passenger vehicles and developed a model based on closing speed and age that the authors noted was in line with Ding et al. While not explicitly stated, given the numbers in GIDAS, we assume only L3 motorcycles were included.

Terranova and Perez (2023) utilized data from MCCS to develop motorcycle injury risk models using principal direction of force (PDOF) as a predictor.

Research objectives

This study expands on existing literature by extending injury risk analysis beyond non-frontal collisions between motorcycles and passenger vehicles. We used biomechanics-based predictors for injury risk modeling and the previously developed concept of *Effective Collision Speed,* which incorporates the influence of impact location on collision dynamics (Schubert et al. 2024). Additionally, we included a variable for potential normal projection, based on vehicle geometry and crash constellation. We examined how incorporating these new predictors affects the modeling of motorcyclist injury risk. The study also explores general patterns of injury risk for motorcyclists involved in crashes with passenger vehicles.

Methods

Data source

This study used GIDAS data on passenger vehicle-motorcyclist collisions. GIDAS is a collaborative initiative between the German Federal Highway Research Institute (BASt) and the German Association for Research in Automobile Technology (FAT) (German In-Depth Accident Study 2025). Since 1999, GIDAS has gathered detailed data on approximately 2,000 traffic crashes annually within the Dresden and Hanover regions and their surrounding areas. For this analysis, we used the GIDAS database as of December 31, 2023, encompassing 42,647 fully documented and reconstructed crashes (FAT et al. 2023).

Inclusion in the GIDAS dataset requires that at least one individual involved in the crash sustained injuries (Unger 2024). Data collection occurs directly at the crash site and involves crucial elements for crash reconstruction, interviews with participants, and thorough documentation of all

injuries, including medical records. Each recorded crash is reconstructed with PC-Crash from the initial critical event to the final positions of the involved participants, with a comprehensive dataset of approximately 3,500 variables per crash. GIDAS offers representative insights into injury crashes in Germany, with survey areas chosen to reflect the country's diverse geography. Data collection adheres to a rigorously defined sampling plan, ensuring a large and statistically representative sample size.

For this study, we focused on cases involving a collision between a motorcycle and a single passenger vehicle, applying specific exclusion criteria outlined in Table 1.

This study considered both L1 and L3 motorcycles (UNECE 2017). Motorcyclists aged 15 years and older were included in the study, as 15 is the legal minimum age for riding smaller mopeds and scooters in Germany.

To align with U.S. categorizations (NHTSA 2024b), passenger vehicles were defined as those weighing under 10,000 lbs and classified as M1 or M2 passenger vehicles or N1 smaller goods vehicles, following the UNECE official definition (UNECE 2017).

Injury severity was assessed using the Maximum Abbreviated Injury Scale (MAIS, 2015 revision of AIS), which captures the most severe injury level sustained by the motorcyclist across all documented injuries (AAAM 2016).

Injury risk model parameters

This study adapts the injury risk modeling framework established for bicycle-passenger vehicle collisions in Schubert et al. (2024) to assess motorcyclist injury risk in collisions with passenger vehicles.

Injury severity was classified using the MAIS level, with injury outcomes modeled as binary variables, indicating whether the motorcyclist sustained at least one injury at the specified AIS level or higher, or was fatally injured. In alignment with ISO 26262 safety standards (ISO 26262 2011), we employed binary logistic regression to estimate the probability of injury at the MAIS2+F, 3+F, 4+F and 5+F levels.

As in our previous work, contacts with the passenger vehicle front were defined to include all contacts forward of the vehicle A-pillar (Schubert et al. 2023; 2024). The motorcycle impact direction was determined based on the vehicle deformation index.

Table 1. Exclusion criteria to extract cases from the GIDAS dataset.

Exclusion criteria
Passengers in sidecars - include only riders and passengers on motorcycle
Motorcycles with trailers
Motorcycle collisions with vehicle trailers
Motorcyclist fell before impact with the car
Motorcyclists who experienced subsequent collisions with other road users (but no exclusion of cases with multiple collision with the same passenger vehicle)
Cases with unknown motorcyclist age, injury, and collision impact locations on the motorcycle and car
Exclude uninjured motorcyclists
Cases with unknown collision speeds and collision angle to determine the relative speed between motorcycle and passenger vehicle
Exclude cases with passenger vehicle slip angles greater than 20°, exclude skidding passenger vehicles
Exclude motorcyclists below age of 15 years

Effective Collision Speed: Speed was modeled using *Effective Collision Speed*, a term designed to capture the impulse experienced by the motorcyclist at impact. This was adapted from the approach developed for cyclists in Schubert et al. (2024) and utilizes the speed of the vehicle and motorcyclist as obtained from the collision reconstructions. In GIDAS, both the motorcycle and passenger vehicle impact speeds (variable VK in GIDAS) are defined. The variable KWINK describes the relative angle between the velocity vectors, which can be used to transform the motorcycle velocity into the passenger vehicle reference frame. From this, relative velocities in the lateral and longitudinal directions can be derived as:

$$v_{lateral} = VK_{bicycle} \cdot \sin\left(KWINK\right) \tag{1}$$

$$v_{longitudinal} = VK_{passenger\ vehicle} - VK_{bicycle} \cdot \cos\left(KWINK\right) \tag{2}$$

$$v_{rel} = \sqrt{v_{lateral}^2 + v_{longitudinal}^2} \tag{3}$$

Given that motorcyclist injury may occur as a result of the initial impulse associated with contact with the vehicle or potential subsequent contacts, the *Effective Collision Speed* term aims to capture the worst-case impulse from these scenarios and is defined as the maximum of three candidate terms:

$$\begin{aligned} &\textit{Effective Collision Speed} \\ &= \max\left(v_{passenger\ vehicle-fric}, VK_{motorcycle}, v_{rel-fric}\right) \end{aligned} \tag{4}$$

where:

- $VK_{motorcycle}$ is the impact speed of the motorcycle.
- $v_{passenger\ vehicle-fric} = A \cdot VK_{passenger\ vehicle}$ represents the friction-adjusted vehicle speed, which captures energy transfer when the vehicle induces rider motion. $VK_{passenger\ vehicle}$ is the impact speed of the passenger vehicle.
- $v_{rel-fric} = \sqrt{v_{lateral}^2 + v_{long-fric}^2} = \sqrt{v_{lateral}^2 + (v_{longitudinal} \cdot A)^2}$ is the friction-adjusted relative speed between the motorcycle and vehicle.

The friction adjustment factor A ($0 \le A \le 1$) scales only the longitudinal component, reflecting that in side contacts, a rider may slide along the vehicle rather than fully engaging, which reduces the effective impulse transfer. In contrast, for frontal and rear impacts full engagement is assumed ($A = 1$).

To determine an appropriate constant value for A, the factor was systematically varied from 0 to 1 in increments of 0.05, and the value yielding the best statistical fit to the injury data was selected. This optimized constant was then applied consistently in all Effective Collision Speed computations for side impacts (see Results section).

For instance, assuming identical impact speeds and collision angle, a motorcyclist colliding laterally with the side of a passenger vehicle yields a lower Effective Collision Speed than when struck from the side by the vehicle's front, where

full engagement with vehicle structures takes place and the frictional reduction does not apply.

As another example, imagine a motorcyclist traveling parallel to a faster passenger vehicle, and the vehicle's side makes contact with the rider. Estimating injury risk based solely on the motorcyclist's own speed or the simple closing speed would underestimate the potential for energy transfer, while using the vehicle's full speed would overestimate it. In such cases, the friction-adjusted vehicle speed and relative speed terms capture the partial engagement more realistically, accounting for both reduced energy transfer and possible post-impact acceleration of the motorcyclist.

Potential Normal Projection: Beyond collision speed, the manner in which the motorcyclist engages with the vehicle may influence injury risk. To capture post-impact motion and the potential for additional injurious contacts, we developed a variable representing the potential for a normal projection (NPROJ) – meaning a perpendicular projection away from the contacting surface – following (Schubert et al. 2024) (Table 2).

Additional Model Predictors: Age is incorporated as a natural cubic spline function to account for non-linear relationships with injury risk. Prior research (Forman and McMurry 2018; Schubert et al. 2023) suggests that young adults have greater biomechanical resilience compared to both younger children and older adults. This approach allows for a continuous representation of appropriate age-related risk rather than categorical segmentation. Sex (female = 1) was modeled to investigate potential physiological and biomechanical differences in injury tolerance.

Helmet use, while a known protective factor, is not included as a predictor of overall injury risk, given the high helmet wearing rate in our dataset and its primary influence on head injuries rather than systemic trauma (Rice et al. 2016).

We explored the possibility of including a variable to distinguish between motorcycles and scooters based on their differing seating positions, but the estimates did not reach statistical significance. A categorical distinction between L1 and L3 motorcycles also failed to produce significant results. Similarly, Airaksinen et al. (2016) found that while motorcyclists tend to suffer slightly more vertebral and midfoot fractures than moped drivers, the overall injury severity - assessed *via* the maximum abbreviated injury scale and median new injury severity scores - did not differ significantly between the two groups.

Model Evaluation: To ensure robustness, we assessed model performance using the areas under the receiver operating characteristic and precision-recall curves (AROC, APRC), applying a 5-fold cross-validation procedure. Additionally, given that injury databases often overrepresent severe crashes (Pfeiffer and Schmidt 2006), we reweighted our models to represent the full spectrum of injurious crashes in Germany. Specifically, weighting factors derived from the 2023 German national data (DESTATIS 2024) were applied, with the Federal Statistical Office of Germany providing distributions specific to motorcycle-passenger vehicle related crashes. Normalized weighting factors were utilized as described in Schubert et al. (2023). Further details on the re-weighting process can be found in the Appendix (supplementary material).

All model predictor variables were selected a priori as outlined above for their potential biomechanical basis.

Results

Descriptive analyses

After applying the outlined exclusion criteria, the resulting dataset for analysis consisted of 2,499 motorcyclist crashes with passenger vehicles. 61.3% of motorcycles were classified as belonging to the L3 category while 38.7% were classified as L1. Of the motorcyclists included in the study, 6% were passengers. Helmet usage was high (86.6%), with unknown helmet status recorded in 10.1% of cases. Notably, only 3.3% of riders were known to be unhelmeted, and helmet wearing rates were similar between male and female motorcyclists.

Protective clothing was less commonly used, with 50.3% of riders not wearing any protective clothing. 24.9% of the motorcyclists wore a complete motorcycle suit, 11.8% wore a motorcycle jacket only, and 1.6% wore only trousers. Protective clothing status was unknown in 11.3% of cases.

The most common crash configurations observed were impacts to the side of the motorcycle by the front of the passenger vehicle (27%) and front-front collisions (25%), where the front of the passenger vehicle is defined up to the A-pillar (Figure 1). Of the 70 side of the motorcycle to rear of the passenger vehicle collisions (2.8%), 12 involved passenger vehicles that were reversing, while in the remaining cases, motorcycles often struck the rear edge of the

Table 2. Definition of the predictor NPROJ.

Potential normal projection – *NPROJ* = 1	No potential normal projection – *NPROJ* = 0
• Front of a taller vehicle (e.g., a van) strikes a motorcycle. • Motorcycle has a frontal impact with the tall portion of the vehicle (e.g., the front wheel of the motorcycle strikes the passenger compartment of a sedan, or any part of a van/SUV/hatchback/wagon behind the A-pillar). • Rear of a tall vehicle strikes a motorcycle while reversing. • Dooring.	• Front of a small vehicle is hit (front defined until A-pillar). • Side or rear of the motorcycle has contact behind the A-pillar of the car with the side of the car. • Motorcycle hits the rear of a sedan. • Side or rear of the motorcycle has contact with the rear of a non-reversing tall vehicle.

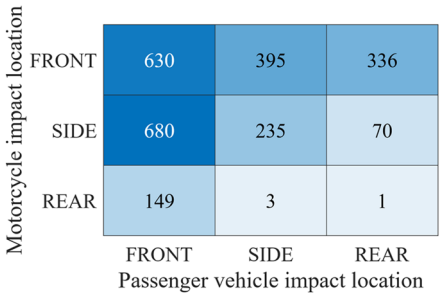

Figure 1. Distribution of impact directions (for passenger vehicles FRONT is defined as impact forward of the A-pillar).

passenger vehicle with their side as they attempted to avoid collision.

The overall median *Effective Collision Speed* was 39.8 kph (Figure 2). For collisions involving the side of the motorcycle, the median *Effective Collision Speed* was approximately 40 kph across impact locations on the passenger vehicle. In contrast, frontal impacts of the motorcycle varied based on where the passenger vehicle was struck. Front-to-front collisions recorded the highest median speed at 45 kph, followed by scenarios where the motorcycle's front hit the side of the passenger vehicle (40 kph), with the lowest speeds observed when the motorcycle's front impacted the rear of the passenger vehicle (27.5 kph). For collisions in which the rear of the motorcycle was struck by the front of a passenger vehicle, the median *Effective Collision Speed* was 35 kph.

Females accounted for only 15.7% of the dataset, with a median age of 27 years compared to 32 years for males (Figure A1, supplementary material). Notably, there was a pronounced peak in the age distribution around 18 years for both sexes with motorcyclist under 18 representing a substantial segment in our dataset (13.7%). Even 21% of all motorcyclists were under 20 years old.

Additionally, 34.4% of the motorcyclists were involved in crash constellations characterized by a potential normal projection. For potential normal projections, the *Effective Collision Speeds* were lower, with a median of 37 kph compared to 40.3 kph, and the 99th percentile speeds were also reduced (118 kph for NPROJ = 1 vs. 130 kph for NPROJ = 0) (Figure A2, supplementary material).

To align with German national statistics, we applied normalized weights with a median of 0.85 (IQR 0.60– 1.30) - i.e., most individual records are up or down-weighted by at most ±40%. After weighting, slight injury cases rise from 64% to 76% and urban crashes fall from 87% to 74%. Longitudinal traffic accidents increase from 22% to 28%, while turning and crossing crashes fall from 63% to 57%. In practice, undersampled crashes i.e., rural settings, slight injuries and longitudinal impacts are up-weighted.

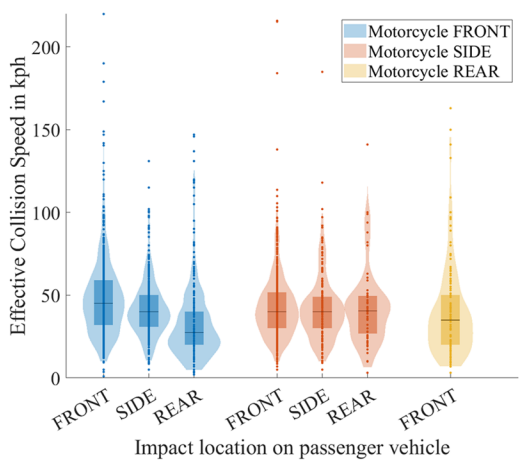

Figure 2. Distribution of the *Effective Collision Speed* depending on the impact direction of the passenger vehicle (FRONT defined as impact forward of the A-pillar) and the motorcycle. Showing the kernel density and a boxplot with the median as solid line and the quartiles.

Injury patterns

In our study, 37% of the motorcyclists sustained moderate or greater (AIS2+) injuries and 11.4% experienced serious or greater (AIS3+) injuries (Table 3). 1.2% of the motorcyclists were fatally injured.

In crashes of MAIS3+ severity, 36.8% were in front-front collisions, followed by 27% involving the front of the passenger vehicle and the side of the powered two-wheeler (PTW), and 15% involved the side of the passenger vehicle and the front of the PTW. Analysis by collision type revealed that for the most common impact scenario - side of the PTW to the front of the passenger vehicle – 11% of collisions resulted in MAIS3+, compared to 17% for front-front collisions and 14% for rear passenger vehicle and side PTW collisions (the latter accounting for only 3.5% of all MAIS3+ cases), indicating that the most frequent crash scenario is not necessarily the most severe.

Common patterns of motorcyclists' injuries are displayed in Figure 3, which shows the body regions with the highest AIS-level injury for each motorcyclist. Overall, the most affected body region was the lower extremities with 52% of all injuries at the maximum AIS-level. Serious or worse injuries (AIS3+) were in 47% to the lower extremities, in 27% to the thorax and in 13% to the head. Among fatally injured motorcyclists, 45% of the most serious injuries involved the thorax, and 38% involved the head.

Over 70% of lower severity (AIS1 or AIS2) injuries were coded as being to the whole area as the injured anatomic structure; contusions (54%) and abrasions (37%) represented most of these injuries (Figure A3, supplementary material).

Table 3. Data sample after applying exclusion criteria.

Total – 2,499		
	Count	*%*
MAIS2+F	926	37.0
MAIS3+F	285	11.4
MAIS4+F	63	2.5
MAIS5+F	36	1.4

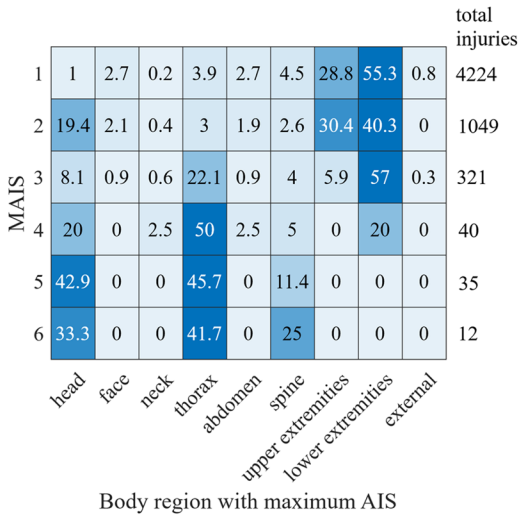

Figure 3. Body region distribution (%) by motorcyclist MAIS level for motorcyclists' most severe injury(ies) (all injuries at each motorcyclist's overall MAIS level are included).

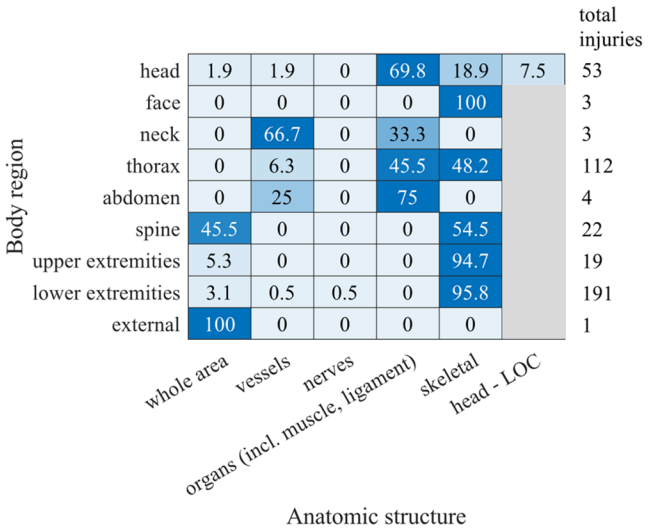

Figure 4. Anatomic structure distribution (%) by body region for motorcyclists' most severe injury(ies) with AIS3+ (all injuries at each motorcyclist's overall MAIS level are included).

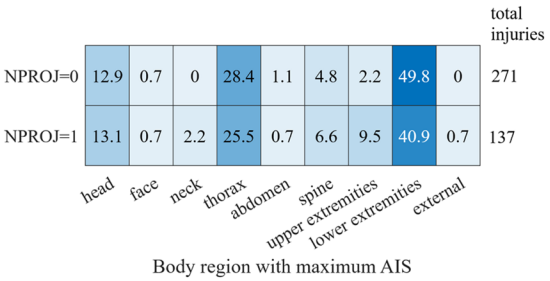

Figure 5. Body region injury distribution (%) by potential normal projection for motorcyclists' most severe injury(ies) with AIS3+ (all injuries at each motorcyclist's overall MAIS level are included).

The head and spine were the only body regions where lower severity injuries to the whole area were not the most common injury. For the head, 80% of lower severity injuries were loss of consciousness (LOC), while for the spine, 85.4% of lower severity injuries were acute muscle strains of the cervical (73%) and thoracic (27%) spine.

For serious and greater (AIS3+) injuries, there was a clear shift in injured anatomic structures toward skeletal (69%) and organ (23%) injuries (Figure 4). Serious head injuries most frequently included brain hemorrhages (40%) and skull fractures (19%). AIS3+ thoracic injuries were primarily characterized by rib fractures (48%) and lung injuries (42%; i.e., contusions, lacerations, and either hemothorax or pneumothorax). Among serious injuries to the lower extremities, femur fractures were most prevalent at 60%, followed by tibia fractures at 28% and pelvic ring fractures at 5%.

We further examined whether variations in motorcyclist interaction with the vehicle - as indicated by NPROJ - were associated with differences in injury patterns. Interestingly, despite lower *Effective Collision Speeds* observed for potential normal projections (Figure A2, supplementary material), the proportion of MAIS3+ injured motorcyclists was similar between cases with a potential normal projection (11.7%) and those without (11.2%) and for both projection modes, 7% of the injuries are AIS3+ injuries. Relative to collisions with the potential for normal projection, a greater proportion of serious thorax and lower extremity injuries were noted for collisions without the potential for normal projection (Figure 5). Serious neck, spine, and upper extremity injuries were more represented in collisions with the potential for normal projection.

We further examined all serious and greater injuries by injury-causing elements (Figure A5, supplementary material). Regardless of potential projection mechanism, the main injury causing part is the outer vehicle structure of the opponent passenger vehicle (~40% of injuries). In cases with the potential for normal projection, a combination of injury-causing elements (22.6%) or contact with the road surface (21.9%) were also commonly observed. These combinations principally consisted of contact with the outer vehicle structure and road surface (48%), the outer vehicle structure and own motorcycle (23%), and the road surface and own motorcycle (13%).

For collisions without the potential for normal projection, in cases with multiple injury causing parts (15%), contact with the outer vehicle structure and road surface (32%), the outer vehicle structure and own motorcycle (15%), and the road surface and environment (15%) were the primary combinations. Those injuries caused by the road surface and environment were only observed for side-to-side collisions. Collisions with the front of the passenger vehicle (75%) represented the majority of cases in which a motorcyclist sustained serious injuries due to contact with the road surface only (28%).

Estimation of friction factor

We applied the same manual maximum likelihood approximation method as outlined in Schubert et al. (2024) for cyclists to estimate the friction factor for motorcyclists that appropriately captures reduced engagement for side contacts. In short, injury risk models (MAIS2+F and MAIS3+F) were fit with *Effective Collision Speed* as the sole predictor, and the friction factor was varied from 0 to 1 in increments of 0.05. For motorcyclists, the maximum log-likelihood was achieved at a friction factor of 0.8 (Figure A6, supplementary material). This result suggests that, assuming a 90-degree collision angle, the contribution of vehicle speed to injury risk in side contacts is reduced by 20% relative to frontal contacts.

Injury risk curves

The detailed estimates and corresponding p-values of the regression models for MAIS2+F and MAIS3+F are provided in Table 4 for the unweighted and weighted dataset. The results for MAIS4+F and MAIS5+F are added in the Appendix (Table A2, supplementary material).

Our analysis consistently demonstrated that an increase in *Effective Collision Speed* is associated with a higher injury risk (Figure 6). This relationship held across all models, underscoring the critical role that collision speed plays in injury severity.

Table 4. MAIS2+F And MAIS3+F motorcyclist injury risk models in collisions with passenger vehicle for unweighted and weighted dataset.

		unweighted			*weighted*		
		Estimate	*Standard Error*	*p-value*	*Estimate*	*Standard Error*	*p-value*
MAIS2+F	*Intercept*	−2,538	0,398	0,000	−3,159	0,430	0,000
	Effective Collision Speed	0,042	0,003	0,000	0,039	0,002	0,000
	Agespline1	−0,162	0,733	0,000	0,570	0,802	0,005
	Agespline2	0,086	0,220		0,271	0,242	
	NPROJ	0,303	0,095	0,001	0,226	0,103	0,028
	SEX Female	−0,067	0,128	0,600	−0,029	0,139	0,837
MAIS3+F	*Intercept*	−3,322	0,596	0,000	−3,832	0,702	0,000
	Effective Collision Speed	0,039	0,003	0,000	0,041	0,003	0,000
	Agespline1	−1,808	1,134	0,000	−1,907	1,354	0,003
	Agespline2	−0,375	0,341		−0,412	0,408	
	NPROJ	0,267	0,143	0,062	0,329	0,167	0,048
	SEX Female	−0,425	0,233	0,068	−0,338	0,265	0,202

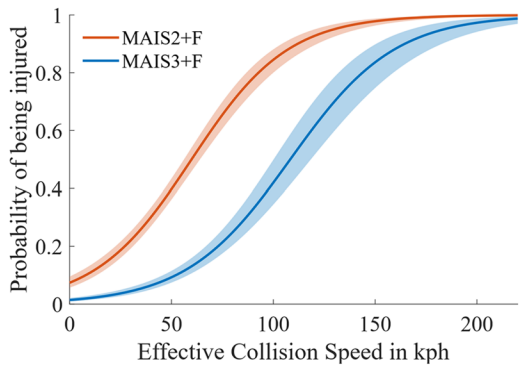

Figure 6. Unweighted injury risk curves for MAIS2+F and MAIS3+F with confidence intervals for NPROJ = 0, age = 32 years (median age in dataset), SEX = male.

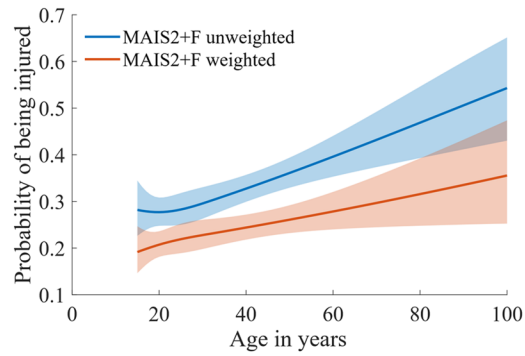

Figure 7. Weighted and unweighted MAIS2+F injury risk curves with confidence intervals for NPROJ = 0, *Effective Collision Speed* = 40 kph (median in dataset), SEX = male.

We observed a higher injury risk for cases with potential normal projection. Notably, the p-values associated with models predicting MAIS3+F were higher than those for MAIS2+F, indicating that the association is less statistically significant for MAIS3+F. Approximately 35% of cases in both the MAIS2+F and MAIS3+F categories exhibited a potential normal projection. This suggests that while the trend is consistent, the strength of evidence varies with the severity threshold.

The models also revealed lower predicted injury risk for female motorcyclists. However, this effect did not reach conventional levels of statistical significance, as evidenced by p-values consistently greater than 0.05. In particular, the MAIS2+F models showed not only high p-values but also high standard errors, which may indicate a lack of precision in these estimates or potential issues related to sample size and variability. Among MAIS2+F cases, only 12.6% involved females, while females accounted for just 8.4% of MAIS3+F cases.

Within sexes, 38.4% of male motorcyclists had an injury at the MAIS2+F severity level compared to 29.8% of females, and 12.4% of males experienced injury at the MAIS3+F severity level versus 6.1% of females. Additionally, median *Effective Collision Speed* was lower for females (34 kph compared to 40 kph for males), and female riders were younger, with a median age of 27 years versus 32 years for males. These factors may contribute to the lower risk predictions for females observed in the regression models.

An interesting divergence emerged when comparing the effect estimates for age between the weighted and unweighted MAIS2+F models (Figure 7). In the weighted model, the risk of injury consistently increased with age, though this increase was not as steep as observed in the unweighted model. In contrast, the unweighted model depicted a slightly decreasing risk from age 15 until around 20, after which the risk began to increase. Above the age of 60 years the confidence intervals widen considerably due to a limited number of cases in this age group.

Overall, the weighted models tend to indicate a lower injury risk (Figure 8), which aligns with the fact that more severe accidents are slightly overrepresented in the GIDAS dataset.

Discussion

This study adapts the cyclist framework introduced by Schubert et al. (2024) for motorcyclists. Expanding on previous research, we evaluated collisions between passenger vehicles and motorcycles and incorporated non-frontal impact events. We examined collisions involving all classes of motorcycles - from traditional motorcycles to mopeds and scooters. Our analysis presents comprehensive injury patterns, and the resulting risk models enable an omni-directional evaluation of injury severity by integrating key biomechanically relevant predictors. Central to the approach is a novel speed parameter that serves as the primary predictor, augmented by motorcyclist age and sex. In addition, we incorporated the NPROJ parameter to assess the likelihood that a collision will result in a normal projection away from the vehicle after the initial impact, thereby capturing the heightened risk associated with subsequent contacts.

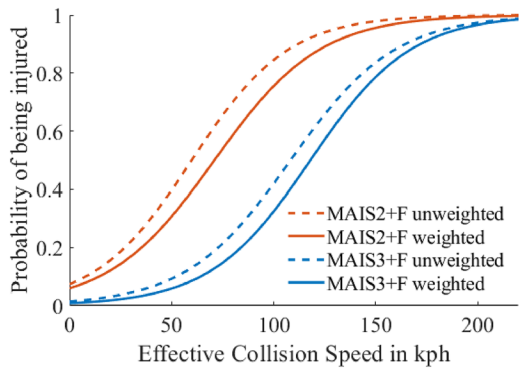

Figure 8. Comparison of injury risk curves for NPROJ = 0, age = 32 years (median age of motorcyclist in dataset), SEX = male.

Applying the bicycle collision methodology with a recalibrated friction factor - rising from 0.7 for bicycles to 0.8 for motorcycles - in the *Effective Collision Speed* framework yielded robust MAIS2 + F and MAIS3 + F models for motorcyclists. The potential normal projection variable proved even more statistically significant than in the original bicycle models, underscoring its heightened relevance in motorcycle collisions and the approach's adaptability to motorcycle crash dynamics.

Comparison to existing injury risk models

In the context of motorcycle-vehicle collisions, the distinction between the striking and struck vehicle, as well as the chosen methodological approach for characterizing impact direction, plays a critical role in understanding injury outcomes. Collision type classifications according to ISO 13232, as used by Otte et al. (2015), consider both the impact location on the motorcycle and the passenger vehicle, along with the collision angle. Ding et al. (2019) analyzed impact locations based on the motorcycle's structural areas (e.g., front, side, or rear). The use of Principal Direction of Force (PDOF) as a predictor variable has been explored by Terranova and Perez (2023). While PDOF indicates the force direction during impact, it does not address the hit structure of both the motorcycle and the opposing vehicle. All these approaches do not capture the biomechanical relevance of post-impact motion, which may contribute significantly to injury severity. To address these limitations, our study incorporated the NPROJ parameter, which captures the potential for a normal projection following initial impact by considering the characteristics of the struck structure. This approach aligns more closely with injury biomechanics, as it reflects the potential trajectory changes that influence secondary impacts with the ground or other objects.

For instance, in cases of rider ejection with potential normal projections, there is a higher likelihood that the rider will contact the passenger vehicle structure during ejection and subsequently be propelled away. This dynamic contributes to a greater occurrence of multiple injury causing part as seen in our analysis. Conversely, in cases of no expected normal projection, a larger proportion of injuries are attributable solely to the road surface and the environment, as the rider may be ejected without contacting the vehicle at all.

We compared the injury risk curves from Lubbe et al. (2022) and Ding et al. (2019) to our models, as reasonably as could be achieved given differences in methodological approaches. Lubbe et al. (2022) focused exclusively on frontal impacts, using closing speed as the primary speed parameter for predicting injury risk. Ding et al. (2019) also employed closing speed in their model. Both studies just considered L3 motorcycles. While these differences complicate direct comparisons, a relative evaluation is still possible by considering our *Effective Collision Speed* alongside the closing speeds used in these studies. For instance, when comparing injury risk estimates, a 10% risk of MAIS3 + F is associated with a closing speed of 50 kph in the Lubbe et al. (2022) model, whereas our weighted model predicts the same injury risk at an *Effective Collision Speed* of 64 kph. Interestingly, our unweighted model aligns more closely, estimating a 10% MAIS3 + F risk at 52 kph. This can be attributed to several factors. First, Lubbe et al. (2022) considers only frontal impacts involving L3 motorcycles, whereas our study includes side impacts and L1 motorcycles (39% of our dataset), which tend to be less severe. This likely results in a higher *Effective Collision Speed* needed to reach the same injury risk threshold. Second, differences in the weighting processes must be noted. Lubbe et al. (2022) weighted their sample by injury severity and accident type for each accident year of the whole German statistics not considering accident location. In contrast, our weighting process applies distributions specific to motorcycle-passenger vehicle related crashes and includes injury severity, accident type, and additionally location (urban/rural/motorway). We regard location as a critical factor because it significantly influences speed distributions and impact dynamics, thus providing a more accurate representation of real-world accident conditions.

When examining the curves presented by Ding et al. (2019), we considered a frontal impact with a passenger vehicle, a stable motorcycle and rider impact. Their model predicts a 10% MAIS3 + F risk at just 22 kph, which is considerably lower than the predictions from both Lubbe et al. (2022) and our models. Notably, the Ding et al. (2019) curves indicate an initial MAIS3 + F risk of 6% even at 0 kph, though the risk increases more gradually compared to the other models. Ding et al. (2019) weighted their dataset only to accident location and injury severity, without accounting for accident type. This difference in methodology may also explain why their results diverge from ours. Ding et al. (2019) predicts an 80% risk of MAIS3 + F at 166kph, compared to 152 kph for our weighted curve, 144 kph for our unweighted curve, and 140 kph in the Lubbe et al. (2022) model. Since weighting adjusts the sample to better match the national distribution, we recommend relying on our weighted model for more representative and generalizable results.

Notably, the age parameter in Lubbe et al. (2022) did not achieve statistical significance for any of the MAIS2 + F, MAIS3 + F or Fatal models. Whereas in our MAIS2 + F and MAIS3 + F models age is statistically significant, in the MAIS4 + F and MAIS5 + F models it is not. Hence using a natural cubic spline to model age and thereby capturing its non-linear relationship with injury risk, appears to offer

advantages for predicting MAIS2+F and MAIS3+F severity levels compared to simply including age. The spline reveals a minimum risk around the age of 21, with injury risk increasing both below and above this age. Given that a substantial proportion of riders in the dataset are under 21, this approach likely improves the model's ability to more accurately reflect injury risk across the full age spectrum, particularly for younger and older motorcyclists. For MAIS4+/5+ levels age does not emerge as a significant factor since *Effective Collision Speed* overwhelmingly governs the accident severity - indicating that at such high impact levels, the role of age becomes negligible.

Comparison of injury patterns: Literature and accident data

In our analysis, we focused on the most severe injury sustained by each motorcyclist (i.e., the maximum AIS-level injury), which may account for differences when compared to studies that include all recorded injuries. Our results show that the lower extremities are the most affected region, representing 52% of all maximum injuries, followed by the upper extremities at 27%, while head and thorax injuries account for 5.2% and 5.4%, respectively.

In MAIDS (ACEM 2009) lower extremity injuries accounted for 31.8% of all reported injuries among motorcycle riders and passengers, followed by upper extremity injuries at 23.9% and head injuries at 18.7%. In contrast, McCarthy et al. (2007) compared data from the MAIDS and UK On The Spot (OTS) studies looking at the most severe injury of each body region, where lower extremity injuries (including the pelvis) accounted for 33% (MAIDS) and 25% (OTS) of injuries, with head injuries at 25% in MAIDS and 18% in OTS, and upper extremity injuries at 23% in both. Furthermore, Airaksinen et al. (2016) found for motorcycle and moped crashes in Finland, upper extremity injuries were most common (40% and 36%, respectively), with lower extremity injuries at around 32–36%. In a complementary analysis by Terranova and Perez (2023) with MCCS data, although lower and upper extremity injuries were frequent overall (32.7% and 24.1%), they contributed less to severe outcomes (15.0% and 1.5% AIS3+ injuries) compared to thorax and head injuries (35.5% and 25.5% AIS3+ injuries). These body regions were similarly the most affected for AIS3+ injuries in our dataset. For AIS2+ injuries, both the ISO13232 and MAIDS databases identified the same three body regions as being most frequently injured (upper extremities, lower extremities, and head), though the proportions varied implying that injury distributions can vary based on severity threshold and study design, even though the overall trends remain consistent (Grassi et al. 2018). A key distinction from these studies is that our analysis exclusively examined motorcyclist injuries resulting from collisions with passenger vehicles. The mentioned studies, as well as the broader GIDAS database, collect data on all collisions involving motorcyclists, including those with fixed objects or other single vehicle collisions. For instance, in the MAIDS project, more than 30% of collisions did not involve contact with a passenger vehicle. Analyzing these collision types collectively may contribute to the observed differences in injury proportions. Additionally, while MAIDS and similar studies were limited in duration, GIDAS has been collecting data continuously since 1999, providing a robust foundation for long-term traffic safety research.

Limitations

The GIDAS dataset presents several inherent limitations. Spanning over two decades, it does not fully reflect the recent advances in passenger car frontal structures designed to protect vulnerable road users - a development that has become increasingly important as Euro NCAP has continuously refined its VRU safety assessments. Moreover, because GIDAS only investigates cases where at least one party is suspected of sustaining an injury, the data are skewed toward more severe collisions. This bias is further compounded by a generally high rate of underreporting of motorcycle accidents, particularly those involving lower injury severities (Amoros et al. 2006; Watson et al. 2015; Airaksinen et al. 2016). Consequently, the models tend to be conservative in their assessment of injury risk.

Our use of *Effective Collision Speed* – supported by the statistical relationship observed for friction factor A - captures collision severity but represents only one method. We also simplified vehicle characteristics: due to missing geometry details (i. e. hood leading edge height, motorcyclist center-of-gravity height), vans were classified as taller for the NPROJ parameter while SUVs were not, due to their heterogeneous front-end designs and the uncertainty regarding whether an individual motorcycle impact occurred above or below the rider's center of gravity. Future work could address this simplified modeling of projection by incorporating a more detailed assessment of the rider's seating position - specifically the center of gravity (COG) - in relation to key structural features of the impact vehicle, such as bonnet height, depth, and angle. Incorporating these parameters may provide a more refined prediction of the likelihood of a normal projection. Additionally, a dedicated injury mechanism study would provide deeper insight into how specific crash constellations influence injury severity. The German vehicle fleet differs significantly from that in the U.S., where larger and taller vehicles are more prevalent, potentially influencing injury risk patterns. Additionally, Euro NCAP features VRU-specific test conditions that promote VRU safety, which are noticeably absent in U.S. NCAP. It would be of interest to explore the transferability of the developed models to the U.S. population by using datasets such as CISS to evaluate their robustness and highlight potential region-specific differences in motorcyclist injury risk.

Helmet use and protective gear are critical for mitigating injuries. Previous studies have shown that helmets significantly reduce head injuries (Otte et al. 2013; Khor et al. 2017; Meng et al. 2023) as well as cervical spine injuries (Crompton et al. 2011). Similarly, protective clothing has been associated with a reduction in hospital admissions and a reduction of soft tissue injuries, though no association was found on the risk of higher severity fracture injuries (De Rome et al. 2011).

In our study, we did not include helmet status or protective gear in the risk models for the following reasons. The 10% of cases with an unknown helmet status would not have been available for modeling. Our dataset showed a high compliance rate (87% known to be helmeted, and only 3% known to be unhelmeted), and in Germany overall, more than 98% of riders wear a safety helmet (BASt and Surges and BASt 2025). In contrast, helmet use in the U.S. varies considerably, especially in states without universal helmet laws. In states that mandate helmet use, 94.3% of riders wear helmets (DOT-compliant and noncompliant), whereas in states without such laws, only 71.9% of riders do so (NHTSA 2024c). Thus, in states without mandatory helmet laws the rates of head injuries might differ. As for protective gear, the variability in quality and design also prevented us from isolating its effects on injury outcomes.

Thus, while helmet use and protective gear significantly reduce injuries to specific body regions, they are not necessarily associated with a reduction in overall injury severity (Rice et al. 2016).

Moreover, regional variations in motorcyclist behavior, protective equipment use, and infrastructure must be acknowledged, as our data are based solely on German crash data. Helmet usage rates, for example, are near universal in Germany, whereas helmet use varies considerably across U.S., particularly those without universal helmet laws. Variations in infrastructure, such as lane configurations and traffic management strategies, could similarly impact the generalizability of these models when applied beyond the German context. Despite these limitations, GIDAS remains the most comprehensive source of in-depth motorcycle crash data currently available, with detailed, reconstructed collision information providing a robust foundation for developing injury risk models.

Significance

This study presents a comprehensive method for assessing motorcyclist injury risk in collisions with passenger vehicles. By integrating biomechanical predictors, including the potential of a normal projection, our models provide an omni-directional evaluation of injury severity across all collision scenarios. Our findings offer concrete guidance for enhancing motorcycle safety through both vehicle- and rider-focused measures. The clear influence of *Effective Collision Speed* and *NPROJ* across all impact angles suggests that vehicle manufacturers could prioritize energy-absorbing front-end structures and optimized bonnet geometries - such as lower leading edges - to mitigate rider projection and reduce impact severity. The derived injury risk models also enable quantitative evaluation of advanced driver-assistance systems (ADAS) within safety assessment frameworks such as Euro NCAP. For instance, by modeling a reduction in *Effective Collision Speed* - e.g. by the intervention of an autonomous emergency braking (AEB) system - we can directly estimate the decrease in MAIS2+F and MAIS3+F injury probabilities. This capability allows safety protocols to be calibrated against real-world risk metrics, demonstrating, for example, how a 10 kph reduction in collision speed translates into a specific percentage drop in severe injury risk and supporting evidence-based enhancement of ADAS performance requirements. During the development of ADAS and automated driving systems, scenario-based testing is employed to evaluate performance across realistic collision situations and these risk functions can help to guide system parameter optimization to achieve the greatest reduction in injury risk across all collision angles and impact directions.

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